MILITARY SPECIFICATION

FILTER ELEMENT SET, CHEMICAL-BIOLOGICAL MASK, M13A2

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a matched pair of filter elements (see 3.8) for chemical-biological masks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation-Packaging, Methods of.
MIL-I-6866	-	Inspection, Penetrant Method of.
MIL-S-43770/1	-	Snap Hooks, Wire Body, Fixed Loop Eye, Flat Spring Closure, With Retainer.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

E5-1-551	-	Masks, Chemical-Biological, Field, M17A1.
E5-1-552	-	Masks, Chemical-Biological, Field, M17A2.
C5-2-1014	-	Valve, Inlet, Chemical-Biological Mask, C20R1.
C26-19-51	-	Machine, Rough Handling, Q113, Assembly.
D5-3-1000	-	Filter Element Set, Chemical-Biological Mask, M13A2.
E136-41-600-13	-	All Purpose Apparatus, Gas Life Testing, Q95, Assembly.
B136-41-850-2	-	Test Apparatus, Agent Gas Sorption, Q216, Assembly.
E136-42-102-20	-	Penetrometer, Filter Testing, DOP, Q127.
B136-44-1-1	-	Resistance Indicator, Inlet and Outlet Valve, Q212, Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components. All materials and components shall conform to the specifications and drawings forming a part of this specification.

3.2 Manufacture. The filter element set shall be manufactured as right and left elements in accordance with Drawing D5-3-1000 and subsidiary drawings listed thereon.

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3.3 Edgeseal configuration.

3.3.1 Butt seals. Butt seals are defined as areas of the edgeseal where there is less than 0.008 inch overlap of the edgeseal material over the filters (items 2 and 5, Dwg D5-3-1000). Butt seals are permitted provided the following conditions are met when examined as specified in 4.4.4.6.1.

(a) Butt seals shall not exceed 1/2 inch in length and shall be firmly bonded to the filter material and screen.

(b) Butt seals shall be separated by a minimum of one inch on the periphery of the element.

(c) The total aggregate length of the butt seals shall not exceed 1-1/2 inches around the periphery of the element.

3.3.2 Screen breakout. There shall be no separation of the screens from the edgeseal.

3.4 Edgeseal strength.

3.4.1 Screen to edgeseal. The bond strength of the edgeseal to the filter element screen shall be no less than an average of 5.0 pounds and no single element shall be less than 2.0 pounds when tested as specified in 4.4.4.6.2.1.

3.4.2 Filter material to edgeseal. The bond strength of the filter material to the edgeseal shall be no less than 18.0 pounds when tested as specified in 4.4.4.6.2.2.

3.4.3 Molded edgeseal strength. The tear strength of the molded edgeseal shall be no less than 18.0 pounds when tested as specified in 4.4.4.6.2.2.

3.5 Connector.

3.5.1 Connector assembly. The connector shall be tightly crimped to the stiffarator and filter material as shown on Drawing D5-3-1000 to effect a leakproof seal. When examined in accordance with 4.4.4.10 there shall be no evidence of cuts or breaks in the surrounding filter material or cracks in the connector itself and the connector shall withstand, without turning, a torque of 6.0 inch-pounds when tested in accordance with 4.4.4.9.

3.5.2 Alternate connector.

3.5.2.1 Functioning. The alternate connector [assembled in a filter element which is installed in an M17A1 Mask (Drawing E5-1-551)] or M17A2 Mask (Drawing E5-1-552) shall withstand 100 installations and removals of the C20R1 Inlet Valve (Drawing C5-2-1014) when tested as specified in 4.4.4.8.1.

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3.5.2.2 Strength. When tested as specified in 4.4.4.8.2, the alternate connector (assembled in a filter element) shall withstand a 225 pound pull or a pull sufficient to separate the connector from the filter element (if such separation occurs at a pull less than 225 pounds) without cracks, shearing, partial shearing or structural failure.

3.6 Dioctylphthalate (DOP) smoke penetration. The DOP smoke penetration of each filter element shall not exceed 0.015 percent before rough handling and shall not exceed 0.025 percent after rough handling (4.4.4.5) when tested as specified in 4.4.4.1, using the DOP smoke at a concentration of 100 to 125 micrograms per liter, an average particle diameter of 0.3 micrometer, and a flow rate of 16 liters per minute.

3.7 Air flow resistance. The airflow resistance of each filter element shall be determined to the nearest millimeter of water and shall be no less than 25 or greater than 45 millimeters of water at an airflow rate of 42.5 liters per minute when tested as specified in 4.4.4.2.

3.8 Matched pairs. The filter elements shall be matched into pairs consisting of a right and left element whose airflow resistances (see 3.7) do not differ by more than 5 millimeters of water. Such a matched pair shall constitute a filter element set.

3.9 Drying. Drying shall be to a level to meet the requirements of 3.10. Under no circumstances shall the elements be heated above 165°F for this purpose.

3.10 Moisture content. The average moisture content of the elements after packaging shall not exceed 0.5 percent, and no individual element shall exceed 1.0 percent moisture, when tested as specified in 4.4.4.3.

3.11 Gas life.

3.11.1 Cyanogen chloride. The cyanogen chloride gas life of the filter element, after equilibration at a temperature of 80° ± 10°F and a relative humidity of 80 ± 3 percent, shall be no less than 7.0 minutes when corrected to a challenge concentration of exactly 4000 milligrams per cubic meter (mg/m^3) when tested as specified in 4.4.4.4. Cyanogen chloride gas life is defined as the time needed to attain an effluent cyanogen chloride concentration of 8 mg/m^3 when the filter element is challenged as follows:

(a) Agent: Cyanogen chloride (CNCl), technical grade
(b) Concentration: 4000 ± 200 mg/m^3
(c) Flow: 25.0 ± 0.8 liter per minute (l/min), intermittent (33-1/2 cycles/min)
(d) Temperature: 70° to 90°F
(e) Relative humidity: 80 ± 3 percent

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3.11.2 Dimethylmethylphosphonate (GB simulant).

3.11.2.1 As received. The dimethylmethylphosphonate (DMMP) gas life of the filter element, as received, shall be no less than 53 minutes when corrected to a challenge concentration of exactly 3000 mg/m^3 when tested (without prior equilibration) as specified in 4.4.4.11.1. "DMMP gas life" is defined as the time needed to attain an effluent DMMP concentration of 0.04 mg/m^3 when the filter element is challenged as follows:

(a) Agent: DMMP, not less than 96.0 percent pure (see 6.6)
(b) Concentration: 3000 ± 200 mg/m^3
(c) Flow: 25.0 ± 0.8 l/min, continuous
(d) Temperature: 75 ± 5°F
(e) Relative humidity: not more than 15 percent

3.11.2.2 After aging. After aging for 120 hours at a temperature of 240°F plus 0° or minus 5°F in accordance with 4.4.4.11.2, the DMMP gas life of the filter element shall be no less than 48 minutes when tested (without prior equilibration) as specified in 4.4.4.11.2. DMMP gas life is defined as the time needed to attain an effluent DMMP concentration of 0.04 mg/m^3 when the filter is challenged as follows:

(a) Agent: DMMP, no less than 96.0 percent pure (see 6.6)
(b) Concentration: 3000 ± 200 mg/m^3
(c) Flow: 25.0 ± 0.8 l/min, continuous
(d) Temperature: 75 ± 5°F
(e) Relative humidity: no more than 15 percent

3.11.2.3 After rough handling. After rough handling for 1 hour in accordance with 4.4.4.11.3, the DMMP gas life of the filter element shall be no less than 40 minutes when tested (without prior equilibration) as specified in 4.4.4.11.3.

3.12 Water absorption. The water absorption of each filter element shall not exceed 5.5 grams, measured in terms of an increase in weight when tested as specified in 4.4.4.7.

3.13 Workmanship. The external surfaces of the filter elements shall be free from foreign matter, imperfection and damage, such as blisters, thin spots, punctures, tears, and pin holes. Thin walls can be detected during external examinations by squeezing the edgeseal. The connector shall be free from ridges, pits, dents, burrs, sharp edges, cracks, or tears (before and after crimping), and other surface imperfections which might impair its use.

3.14 Preproduction. Prior to the start of regular production, and prior to implementation of any major change (see 6.5), a preproduction lot of filter elements shall be produced for examination and testing in accordance with 4.3. If the alternate design connector (Drawing D5-3-1002) is used, a preproduction lot of connectors shall be produced for examination and testing in accordance with 4.3.1.2 and 4.3.2.2.

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3.14.1 Mold qualification. After preproduction has been completed and approved, the introduction of any new or rehabilitated molds shall require mold qualification in accordance with 4.3.

3.15 Control of manufacture. The contractor shall provide the Government with the exact (not typical) procedures, process controls and conditions and their limits used to produce filter elements under this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.15, and section 5 for which specific inspection has not been provided in this specification have been satisfied. In addition, the contractor shall provide objective evidence that the same methods, materials, equipment, and processes are used during regular production as were used in the manufacture of the preproduction sample (see 4.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample.

4.3.1.1 Filter element. A preproduction lot of 1000 matched pairs (see 3.8) of filter elements or 10 elements from each edge sealing mold cavity, whichever is greater, shall be produced and packaged using the same methods, materials, and equipment as will be used during regular production.

4.3.1.2 Alternate connector. If the alternate connector is used, an additional preproduction sample of 105 (52 right and 53 left) filter elements with the alternate connector shall be produced using the same methods, materials, and equipment as will be used during regular production.

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4.3.1.3 Mold qualification. One hundred elements or 10 elements from each edgesealing mold cavity, whichever is greater, shall be produced and packaged using the same methods and equipment as will be used in regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Each filter element of the preproduction lot shall be inspected and tested for all of the characteristics included in the classification of defects (see 4.4.3.4).

4.3.2.2 For destructive tests. Two hundred seventy-five matched pairs (see 3.8) shall be taken at random from the packaged (bagged) preproduction lot of filter elements and tested as specified in table I. If the alternate connector (4.3.1.2) is used, the 105-element sample shall be tested in accordance with 4.4.4.8.

4.3.2.3 Mold qualification test. Testing shall be conducted as specified in table III.

4.3.3 Acceptance/rejection criteria. The sample filter elements shall meet all the examination and tests specified by 4.3.2.1 and the acceptance/rejection criteria specified in table I (4.3.2.2) to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production. Acceptance/rejection criteria for the mold qualification test shall be as specified in table III.

4.4 Quality conformance inspection.

4.4.1 Lots and sub-lots.

4.4.1.1 Lotting. A lot shall consist of the packaged filter elements produced by one manufacturer, from the same materials, and under essentially the same manufacturing conditions. However, no more than one lot of filter material or edgeseal material shall be represented in any one lot of finished filter elements.

4.4.1.2 Sub-lotting. For the purpose of the bag leakage and heat-sealed seam tests of 4.4.4.12.1 and 4.4.4.12.2 respectively, a sub-lot shall be defined as follows:

The sublot shall contain no more than one regular production lot of filter elements and shall consist of consecutively packaged sets.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling of filter element pairs shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Sampling of filter elements shall be conducted in accordance with table II. Sampling shall be in matched pairs.

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TABLE I. Preproduction testing

Characteristic	Sample	Requirement	Test	Accept/reject criteria Acc	Accept/reject criteria Rej	Remarks
1. Bag leakage and heat-sealed seam test (inner and outer bags)	25 pairs	5.2	4.4.4.12.1 4.4.4.12.2	0	1	
2. Moisture content	50 pairs	3.10	4.4.4.3	0	1	
3. Water absorption	25 pairs	3.12	4.4.4.7	0	1	Same sample as used for 1 above
4. Edgeseal	50 pairs			1	2	Same sample as used for 2 above shall be used for 4 a and b.
a. Thickness		Dwg D5-3-1000 (see Sec B-B)	4.4.4.6.1			
b. Butt seals		3.3.1	4.4.4.6.1			
5. Edgeseal strength				0	1	
a. Screen to edgeseal	50 elements	3.4.1	4.4.4.6.2.1			Same sample as used for 7 below. 5b and 5c are the same sample and these tests are performed simultaneously
b. Filter material to edgeseal	50 elements	3.4.2	4.4.4.6.2.2	0	1	
c. Molded edgeseal strength		3.4.3	4.4.4.6.2.2			
6. DOP smoke penetration after rough handling	50 pairs	3.6	4.4.4.5 4.4.4.1	0	1	
7. Connector torque	50 pairs	3.5.1	4.4.4.9	0	1	
8. Connector assembly	50 pairs	3.5.1	4.4.4.10	0	1	Same sample as used for 2 above

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TABLE I. Preproduction testing (continued)

Characteristic	Sample	Requirement	Test	Accept/reject criteria Acc	Accept/reject criteria Rej	Remarks
9. Cyanogen chloride gas life	25 pairs	3.11.1	4.4.4.4	0	1	
10. DMMP gas life						
a. As received	25 pairs	3.11.2.1	4.4.4.11.1	0	1	
b. After aging	25 pairs	3.11.2.2	4.4.4.11.2	0	1	
c. After rough handling	25 pairs	3.11.2.3	4.4.4.11.3	0	1	

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TABLE II. Regular production testing

Characteristic	Requirement	Test	Sample size	Accept/reject criteria
1. Moisture content and connector assembly [1]	3.10 3.5.1	4.4.4.3 4.4.4.10	MIL-STD-105 Level S-3	AQL 0.65% defective
2. Edgeseal [2]			MIL-STD-105 Level S-3	AQL 2.5% defective
a. Thickness	Dwg D5-3-1000 (See sec B-B)	4.4.4.6.1		
b. Butt seals	3.3.1	4.4.4.6.1		
3. Edgeseal strength			MIL-STD-105, level S-4	AQL 0.25% defective
a. Screen to edgeseal strength	3.4.1	4.4.4.6.2.1	1 from each pair	
b. Filter material to edgeseal strength and molded edgeseal strength [3]	3.4.2 3.4.3	4.4.4.6.2.2	1 from each pair	
4. DOP smoke penetration after rough handling	3.6	4.4.4.5 4.4.4.1	Double sampling plan 1st 80 elements 2nd 80 elements	Accept, Reject 1 4 4 5
5. Cyanogen chloride gas life	3.11.1	4.4.4.4	MIL-STD-105, level S-2	AQL 1.0% defective
6. DMMP gas life				No defects permitted
a. As received	3.11.2.1	4.4.4.11.1	5 each	
b. After aging	3.11.2.2	4.4.4.11.2	5 each	
c. After rough handling	3.11.2.3	4.4.4.11.3	16 each	
7. Bag leakage and heat-sealed seam test (inner and outer bags)	5.2	4.4.4.12.1 4.4.4.12.2	MIL-STD-105 Level S-4 for each bag	AQL 1.0% defective
8. Water absorption [2]	3.12	4.4.4.7	MIL-STD-105, level S-4 but no less than 50 pairs	AQL 1.0% defective (Each defective element is considered as defective pairs)
9. Connector torque [2]	3.5.1	4.4.4.9	MIL-STD-105, level S-3	AQL 0.65% defective

NOTES: [1]. Connector assembly test shall be performed after moisture content test on same sample.
[2]. Edgeseal test (thickness and butt seals) shall be performed after water absorption and connector torque tests on same sample.
[3]. Molded edgeseal strength and filter material to edgeseal strength tests are performed simultaneously on the same sample. In the event rupture of the filter material occurs prior to completion of the test, replace the sample element(s) and re-run the test.

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TABLE III. New mold qualifications

Characteristic	Sample	Requirement	Test	Criteria Acc	Criteria Rej	Remarks
1. DOP smoke penetration	100 elements	3.6	4.4.4.1	0	1	
2. DOP smoke penetration after rough handling	25 elements	3.6	4.4.4.5 4.4.4.1	0	1	Same sample as used for 1 above.
3. Edgeseal a. Thickness	25 elements	D5-3-1000 (See sec B-B)	4.4.4.6.1	0	1	Same sample as used for 1 above.
b. Butt seals		3.3.1	4.4.4.6.1	0	1	
4. Edgeseal strength a. Screen to edgeseal	10 elements	3.4.1	4.4.4.6.2.1	0	1	Same sample as used for 3 above. 4b and 4c performed simultaneously.
b. Filter material to edgeseal		3.4.2	4.4.4.6.2.2	0	1	
c. Molded edgeseal strength		3.4.3	4.4.4.6.2.2	0	1	

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4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample pairs of filter elements from the regular production lots shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For destructive tests. Sample matched pairs of filter elements shall be taken at random from the completed, packaged (bagged) regular production lots and tested as specified in table II, except for the sample required for DMMP gas life which shall be selected at random from the production line and tested as specified in table II.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Filter element set, chemical-biological mask, M13A2 (Drawing D5-3-1000).

Categories	Defects	Acceptance standards
Critical:		
1	DOP smoke penetration excessive	4.4.4.1
2	Airflow resistance incorrect	4.4.4.2
3	Filter element pairs mismatched (see 3.8) (check immediately prior to heat sealing of bag)	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Plan view profile incorrect	
103	Connector loose or incorrectly located	
*104	Diameter (1.253) incorrect	
105	Diameter (1.535) incorrect	
106	Height of nozzle (0.268)	
107	Workmanship (see 3.13) substandard	
108	Screen breakout (3.3.5)	

*Reference Drawing D5-3-1002

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Minor: AQL 2.5 percent defective

201	Spherical radius incorrect
202	Protective finish missing or incorrect
203	Overall length incorrect
204	Angle (15° 30' ± 30') incorrect
205	Marking missing, illegible, or incorrect

(b) Packaging (section 5)

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Marking missing or incorrect

4.4.4 Tests.

4.4.4.1 DOP smoke penetration. DOP smoke penetration of the filter elements shall be determined with the Q127 Filter Penetrometer (E136-42-102-20).

4.4.4.2 Airflow resistance. Airflow resistance of the filter elements shall be determined with the Q212 Inlet and Outlet Valve Resistance Indicator (B136-44-1-1).

4.4.4.3 Moisture content. Moisture content of the filter elements shall be determined as follows:

Weigh each sample element to the nearest 0.1 gram (wet weight) immediately after unpackaging. Place the elements in a forced-air drying oven at 160° ± 5°F and a maximum of 5 percent relative humidity until constant dry weight, within 0.1 gram, is reached. Record the final weighing (dry weight), to the nearest 0.1 gram. Calculate the percent moisture as follows:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{\text{wet weight}}$$

4.4.4.4 Cyanogen chloride gas life. The filter elements shall be equilibrated to the specified relative humidity at the required temperature for the required period of time and then tested for cyanogen chloride gas life of the filter element singly using the Q95 All Purpose Gas Life Testing Apparatus (E136-41-600-13).

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4.4.4.5 Rough handling. The sample filter elements shall be rough handled, eight (4 pairs) at one time, for 60 ± 5 minutes in the Q113 Rough Handling Machine (C26-19-51), together with a 1.2 ± 0.2 pound rubber ball of 3-5/8 ± 3/32 inch diameter. The hardness of the rubber ball shall be 60 ± 10 Shore Type A. A rubber protector shall be used to prevent damage to the connector.

4.4.4.6 Edgeseal.

4.4.4.6.1 Edgeseal thickness and configuration. The sample filter elements shall be sectionalized, using a steel rule die or a similar cutting tool, in such a manner that each cut section shall be 3/4 ± 1/16 inch wide with a 1/2 inch minimum depth. The cut edges shall be at right angles to the outer wall of the edgeseal. A minimum of three sections shall be cut from each filter element. The cut edges shall be clean and straight. Sections shall be cut along each edge and examined for thickness and butt seals.

4.4.4.6.2 Edgeseal strength.

4.4.4.6.2.1 Screen to edgeseal. The bond strength between the screen and the edgeseal shall be determined by die cutting a section 1 inch wide by approximately 1-1/2 inch long from the long flat edge of the filter element. The outer screen shall be separated on each side to form two tabs capable of being placed into the jaws of a tension testing machine. The jaws of the testing machine shall be no less than 2 inches wide. The machine shall be fitted with maximum load indication. A straight axial pull shall be applied at the rate of 2 ± 1/8 inch per minute until failure occurs.

4.4.4.6.2.2 Filter material to edgeseal and molded edgeseal strength. The bond strength between the filter material and the edgeseal, and the strength of the molded edgeseal shall be determined by die cutting a section 1 inch wide by approximately 1-1/2 inches long from the long flat edge of the filter element. The section of the stiffarator shall be removed and the two filter pads shall be carefully separated so as to form two tabs capable of being placed into the jaws of a tension testing machine; no screen shall be included in the tab. The jaws of the testing machine shall be no less than 2 inches wide. The machine shall be fitted with maximum load indication. A straight axial load shall be applied at the rate of 2 ± 1/8 inch per minute until failure occurs. The separation of the filter material from the edgeseal or rupture of the edgeseal prior to meeting the requirements of 3.4.2 or 3.4.3 shall be considered a failure.

4.4.4.7 Water absorption. Water absorption of the filter elements shall be determined as follows:

(a) Weigh each of the sample filter elements to the nearest 0.1 gram and record the "dry" weight.

(b) Stand the elements upright on their long flat edge in a suitable dish, and add room temperature distilled water to the dish until the water is 1 ± 1/16 inch deep.

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(c) Allow the elements to stand in the water for 30 ± 1 minutes.

(d) At the end of 30 minutes, remove the elements from the dish of water and remove all visible surface moisture with a cloth or towel. The elements should appear to be dry.

(e) Within 10 minutes after removal from the water, weigh each of the sample elements again to the nearest 0.1 gram and record the "wet" weight for each of the sample elements to obtain the water absorption.

4.4.4.8 Alternate connector testing (for preproduction only).

4.4.4.8.1 Functioning. Fifteen filter elements of the sample incorporating the alternate connector shall be assembled into M17, M17A1 or M17A2 Masks. Inlet valves shall be installed and removed from each connector at least 100 times. More than one inlet valve per connector may be used. At the completion of the test, the inlet valve shall snap on the connector, be securely attached to the connector, and be removable from the connector by the use of a snap hook conforming to part number M43770/1-SWZE3 of MIL-S-43770/1.

4.4.4.8.2 Strength. Five of the 15 elements used in 4.4.4.8.1 and 30 of the preproduction samples shall be conditioned in groups of 35 each for at least 48 hours at the following temperatures: the first group at 72° ± 2°F, the second group at 150° ± 2°F, and the third group at -60° ± 5°F. Each alternate connector shall be mounted in a tensile testing machine and pulled at a rate of travel of 2 inches per minute to a load of at least 225 pounds or until the connector separates from the filter element (failure of filter material). The connector shall be held at three points on the top flange simulating an inlet valve; the filter pad shall be held in a nesting device and a straight pull applied. At the completion of pull test each connector shall be inspected for damage such as distortion, cracks, shearing, partial shearing, or structural failure to meet the drawing requirements. After this test each connector shall be tested in accordance with method B, type I or II, MIL-I-6866. The connector shall be sectionalized with a jeweler's saw or high speed cutting device to avoid damaging the connector and examined through an optical comparator with at least 20 X 1 magnification to determine if the connector meets the requirement of 3.5.2.2.

4.4.4.9 Connector torque test. The connectors of filter elements taken from samples for destructive testing shall be tested on a contractor-designed torque tester.

4.4.4.10 Connector assembly. Filter elements shall be cut longitudinally at the junction of the top and bottom filters, completely around the periphery through the edgeseal wall so that the top filter can be separated from the bottom filter. The connector shall be carefully removed by sectionalizing without disturbing the filter material. The connector shall be sectionalized with a jeweler's saw or high speed cutting device to avoid damaging the connector. The screens shall be cut away. The top and bottom backing layers shall be examined for cuts or other damage, and then carefully peeled away. The core layers shall be examined for cuts or "gasketing" or other damage by grasping

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the edge of the core layers with thumb and forefinger at the opening for the connector and lightly tugging at the core layers in the same horizontal plane. Care must be exercised that no bending action is performed during this tactual examination. The sectionalized connector shall be examined for the requirement of 3.5.1.

4.4.4.11 DMMP (GB simulant) gas life.

4.4.4.11.1 As received. The DMMP gas life of the filter elements shall be determined singly using the Q216 Agent Gas Sorption Test Apparatus Assembly (B136-41-850-2) using DMMP (see 6.6).

4.4.4.11.2 After aging. Aging of the filter elements shall be accomplished by storing them in a forced air circulating oven for the required period of time at the specified temperature. Prior to aging, the filter elements shall be removed from the sealed bags and placed on their long flat edge on wire racks in the oven for the duration of the aging period. After aging, the elements shall be tested for DMMP gas life as specified in 4.4.4.11.1.

4.4.4.11.3 After rough handling. After rough handling the filter elements as specified in 4.4.4.5 for the required period of time, the elements shall be tested for DMMP gas life as specified in 4.4.4.11.1.

4.4.4.12 Bag leakage and heat seal test.

4.4.4.12.1 Bag leakage test. Both the liner and outer bag of each unit shall be tested as specified in MIL-P-116 except that the Hot water technique shall be used and after removal from the water, the specimens shall be thoroughly dried, then cut open and examined for any moisture inside the bag. Any moisture inside the bag shall be evidence of leakage.

4.4.4.12.2 Heat seal test. The heat seal test shall be performed in accordance with MIL-P-116.

4.4.5 Acceptance/rejection criteria. The filter elements shall meet the acceptance/rejection criteria specified in 4.4.3.4 and table II.

5. PACKAGING

5.1 Unit packing, packing, and marking, interplant shipment (see 6.4). A uniform quantity of filter element sets, unit packed as specified in 5.2, shall be unit packed and packed to prevent damage in shipment from the supply source to the first receiving activity. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Exterior shipping containers shall be marked in accordance with Fed. Std. No. 123 and shall include contract number.

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5.2 Repair parts. When filter element sets are procured for issue and storage as a repair part, preservation, unit packing, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. This specification covers filter element sets intended for to be used in the M17, M17A1 and M17A2 Field Chemical-Biological Masks.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Precautions. Care should be exercised during cutting and handling of the filter material to avoid punctures, cuts, surface abrasions, and loss of charcoal from the edges, and to minimize moisture pickup during the entire manufacturing process.

6.4 Interplant shipment. Packaging for interplant shipment is for supplies and materials that do not directly enter the military supply system. Typical interplant shipments are shipments from a vendor to a contractor or from a contractor to a military arsenal.

6.5 Major change. The occurance of one or more of the following conditions shall be regarded as evidence of a major change in materials, processes, or equipment and shall constitute cause for a new preproduction lot:

(a) Changes to edgeseal material beyond the requirements of 3.1 of MIL-E-51420 or change of contractor of material.
(b) Change in screening or contractor of material.
(c) Change in cure temperature beyond limits of approved preproduction sample.
(d) Decrease in cure time from that of approved preproduction sample.
(e) Increase in preheat temperature above that of approved preproduction sample.
(f) Change in mold design which will affect the filter element configuration as shown on Drawing D5-3-1000.

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6.6 Dimethylmethylphosphonate (DMMP) test reagent. DMMP may be purchased from Mobil Chemical Company, Industrial Chemicals Division, 801 E. Main Street, P.O. Box 1136, Richmond, VA 23208; or Virginia Chemicals, Inc., 3340 W. Norfolk Road, Portsmouth, VA 23703.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-51425A(EA)	FILTER ELEMENT SET, CHEMICAL-BIOLOGICAL MASK, M13A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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